

When nobody answers the tone.

Rural ambulance services are not short of *trucks*. They are short of **crews** — and no one measures when a crew fails to form. We do, from the agency's own dispatch records, and we turn it into the funding case they can't write themselves.

Team [your team name] Category Rural health infrastructure ↓ scroll or press →

Everyone is optimizing the wrong decision.

EMS analytics is a mature, crowded market — and every product in it is built for a city. They answer one question: **where should we position our ambulances?**

That question requires having ambulances to position. It assumes a fleet, a dispatch center, staffed stations, and an analyst to read the dashboard. Vendors compete on response-time reporting for agencies that already run dozens of units.

A county with **one ambulance** has no placement decision to make. There is one truck and one place it can be. Every dollar of that industry's R&D is spent on a question these agencies cannot ask.

Competing in that ocean means selling a fleet-optimization tool to agencies with no fleet.

— why we stopped building a map

Two thousand agencies nobody sells to.

Not because they don't need software — because the industry decided they can't pay for it.

4.5M Americans live more than 25 minutes from an ambulance station. 4 of 5 rural counties contain such an area. Univ. of Southern Maine / FORHP, 41 states	–25% volunteer firefighters since 1984, while call volume roughly tripled. Volunteers provide 90%+ of rural EMS. NVFC · NRHA workforce research	63% of one state's volunteer-staffed EMS agencies do not provide continuous 24/7/365 coverage. Wisconsin EMS reporting	1 in 3 rural volunteer agencies are at risk of closing outright — they cannot cover operating costs. National Rural Health Association
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These agencies are invisible to the market for a structural reason: in many states EMS is not a legally essential service, so no one is required to fund it. They lose an average of **\$1,526 on every transport**. There is no software line in the budget, because there is barely a budget.

The constraint isn't space. It's whether anyone shows up.

Urban EMS fails when every unit is already on a call. Rural EMS fails a different way — the tone goes out and **no crew forms at all**.

WHAT THE INDUSTRY MODELS

Unit unavailable

The ambulance is committed to another call. Classic queueing theory. Real in cities, where units run busy 30–50% of the time.

Rural utilization runs 5–15%. Simultaneous-call collisions are rare by construction. This is not the binding constraint.

WHAT ACTUALLY HAPPENS

No crew assembles

70–74% of rural EMS providers are not full-time EMS — they hold outside jobs. At 10am on a Tuesday, the roster is at work.

55% of rural agencies have six or fewer people providing 80% of staffing. When two of six are unreachable, the truck doesn't roll.

This failure has **structure** — it clusters by hour of day and day of week, because employment does. That makes it measurable, and therefore manageable. Nobody measures it.

What we removed is why we can serve them.

ELIMINATE

- **Unit-placement optimization.** There is no placement decision with one ambulance. Removing it removes most of the engineering cost of an EMS analytics product.
- **Lives-saved / value-of-life claims.** The evidence doesn't support them, and removing them removes our need to defend them.
- **Onboarding, integrations, training.** No install, no login, no IT project.

REDUCE

- **Price to zero at entry.** The first analysis is free — the agency has no budget to spend, so we don't ask for one.
- **Data burden.** One CSV export, any format. Column names are matched automatically.
- **Time to value.** Minutes, not a deployment.

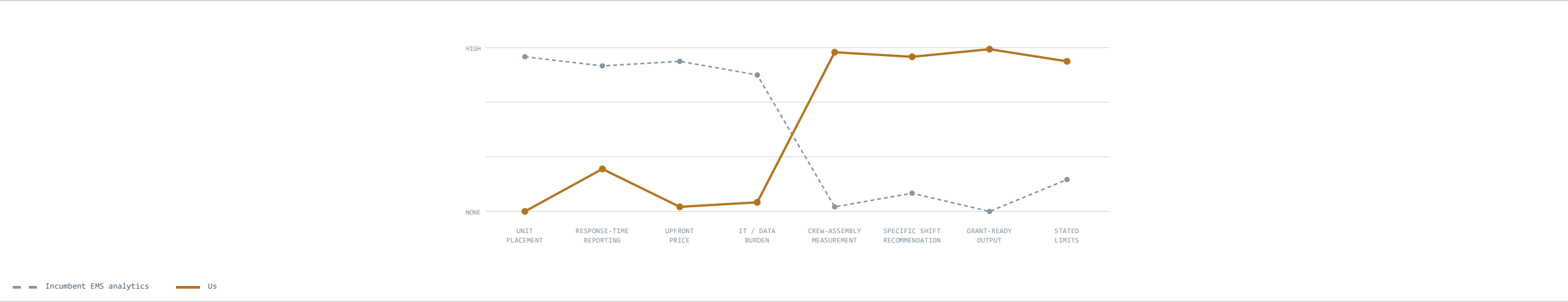
RAISE

- **Specificity.** Not "you're understaffed" but "Tue–Thu 09:00–15:00, 20 hours, covers 61% of your failures, \$62,400/yr."
- **Stated limitations.** Every output names what it does not claim. In a sector burned by vendor overpromising, this is a feature.
- **Privacy.** Patient columns are stripped before analysis, and the report says which.

CREATE

- **Crew-assembly measurement.** A metric nobody in EMS currently reports — how often no crew formed, and when.
- **Grant-ready evidence as the product.** The output is formatted as the sections a funder asks for.
- **A buyer who isn't the user.** Grant programs and regional agencies pay; the volunteer service uses.

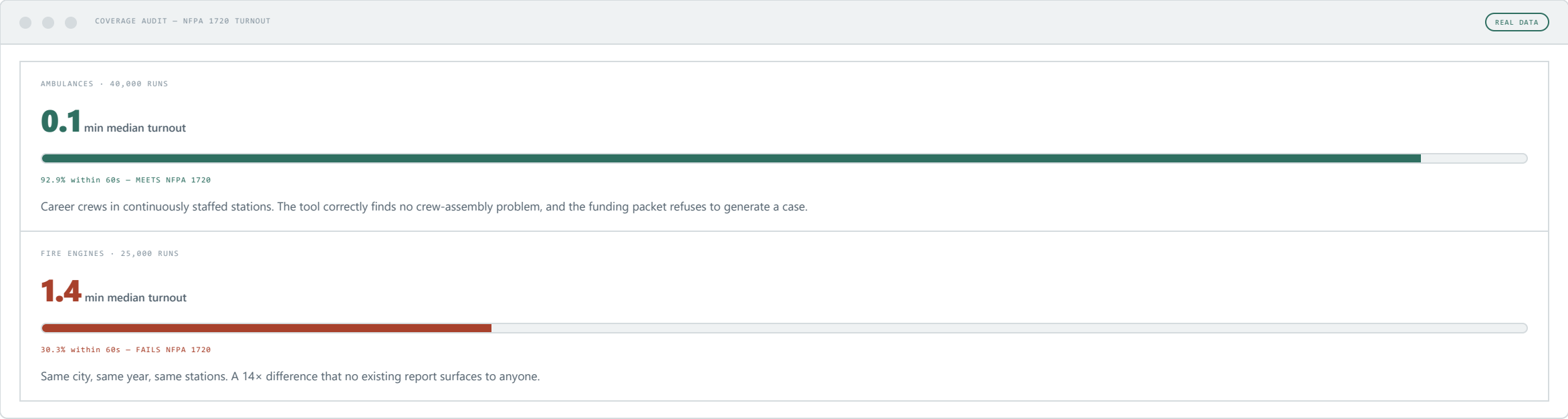
We compete on none of the same factors.



Left four factors: what the industry competes on, and what we eliminated or reduced to reach an unserved buyer. Right four: what we created, where no incumbent currently scores.

The same metric, two real departments.

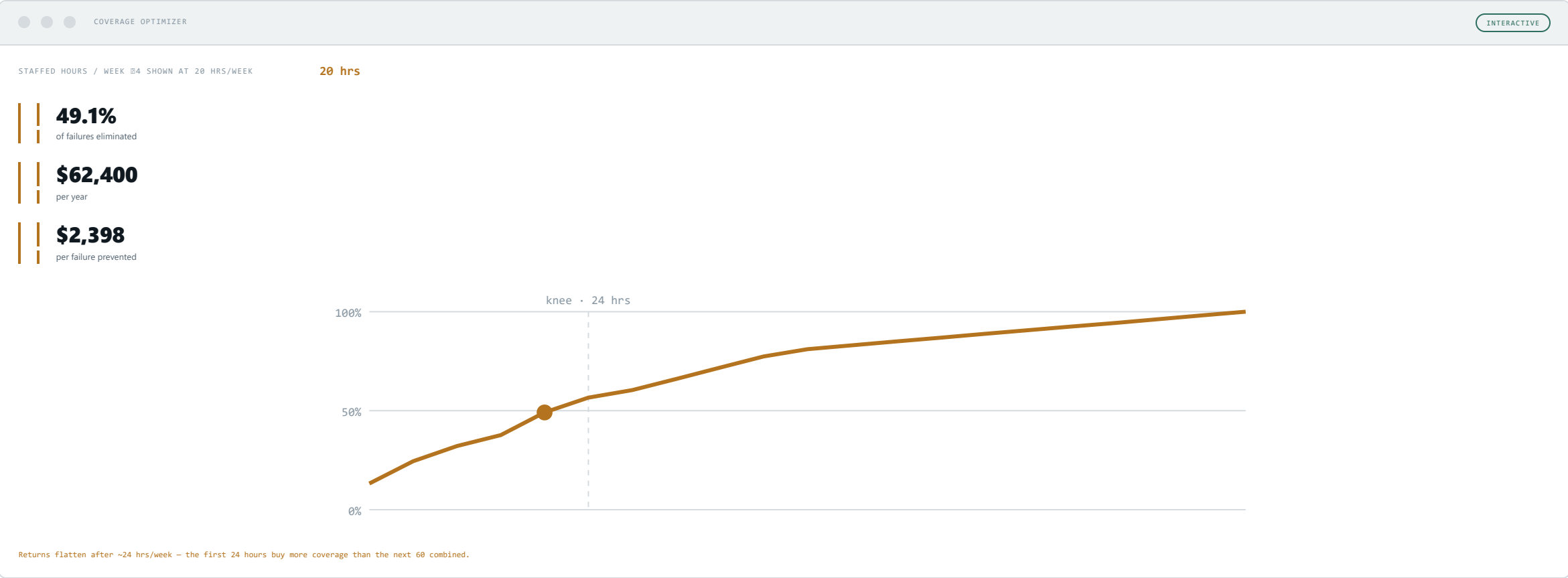
Both datasets are real San Francisco dispatch records from 2024, run through the tool unmodified. The measurement nobody currently reports is the gap between the tone going out and the wheels turning.



If a **well-funded career department** misses the national turnout standard on 70% of its runs and nobody noticed, the question for **24,208 volunteer departments** answers itself.

Which hours should we actually buy?

Failures spread across 168 hour-of-week slots. This is the real returns curve computed from that 40,000-run dataset — not a mockup.



Real output from the shipped tool. Coverage is bought in contiguous blocks of 4+ hours because nobody staffs scattered single hours, and a duty crew is credited with eliminating crew-assembly failures inside its window – not simultaneous-call collisions, which need a second unit rather than more staffing.

We tested it on an agency that doesn't have the problem.

40,000 real San Francisco Fire Department ambulance dispatches, calendar year 2024.
SF runs career crews from continuously staffed stations — so a correct tool should find **nothing wrong**.

0.1_{min} median tone-out to en route DataSF · dataset nuek-vuh3	0.13% of calls flagged as a crew-assembly failure 40,000 incidents analyzed	1.5× weekday-daytime ratio — essentially flat, as expected for career staffing no volunteer pattern present
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It reported no problem, and the funding packet **refused to generate a case**, printing "not a funding case" instead. A tool that manufactures findings on a well-run agency would discredit every legitimate report it ever produces.

That is the only claim this validation supports: **the method doesn't invent problems**. Proving it detects them requires a rural volunteer agency's data — which is the ask on the final slide.

We falsified part of our own thesis.

Before building, we ran three experiments on our core statistical claim — that pooling data across counties unlocks prediction at rural volumes. **It mostly doesn't.**

DISPROVED	"Rural counties can't model themselves." False above ~200 calls/year with a few years of history. Pooling added 0.2–1.1% — not the unlock we assumed.
DISPROVED	"Faster response saves lives." A systematic review of 115 studies and 691,056 patients found insufficient evidence at the margins we operate in. We removed every outcome claim and compute no value-of-life figure anywhere.
DISPROVED	"The gap is the ambulance being busy." Rural utilization is too low for that. We rebuilt the model around crew assembly — the critique that most improved the product.
SURVIVED	The workforce failure is real, structured, and unmeasured. Documented across multiple states, patterned by employment, and reported by no existing tool.
SURVIVED	Prevention outreach reduces emergency utilization. 24–63% reductions in 911 use across published programs, one of them a randomized controlled trial.

Every figure in this deck is sourced. Where our own data is modeled rather than observed, the product labels it as modeled – on the screen, not in a footnote.

The user has no money. The buyer does.

Prevention savings land on Medicaid and hospitals. Coverage costs land on an agency losing \$1,526 a transport. That mismatch is why this market looks unserveable — and why the entry point is free.

BEACHHEAD
Regional EMS agencies
One relationship covers 5–10 counties. They hold budgets, staff who read data, and a mandate for exactly this. California alone has ~33.
BEST-ALIGNED
Hospital-run services
Systems that operate the ambulance feel both pockets — the ED cost and the coverage cost. Already common in rural counties.
FUNDING MECHANISM
Federal & state grants
SIREN funds rural EMS recruitment, retention and technology at up to \$200K per award. Applications require documented need — our output.

We don't sell against their budget. **We help them create one**, then get paid from it. The free audit is customer acquisition, product validation, and grant groundwork in a single motion.

Rural EMS is the beachhead. Volunteer fire is the market.

We are not an ambulance company. We measure whether a **distributed, on-call, volunteer workforce actually assembles when summoned**. Ambulances were where we found the problem. They are not where most of it lives.

24,208 U.S. fire departments that are all- or mostly-volunteer — 82% of every department in the country NFPA / NVFC fire service census	13× the buyer universe of rural EMS counties, reachable with zero product change 24,208 vs ~1,800	676,900 volunteer firefighters — the lowest number ever recorded, down 6% in a single year NFPA, most recent census
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WHY IT'S NOT A PIVOT

Identical data, identical failure

Fire dispatch exports carry the same two timestamps we already parse. The failure is the same: the tone goes out, and the question is whether a crew forms.

In most rural counties **the fire department is the EMS agency**. Same buyer, wider door.

WHY THEY MUST CARE

NFPA 1720 exists for this

The national deployment standard for volunteer departments exists precisely because, in its own framing, volunteers cannot guarantee availability the way on-duty career staff can.

It asks departments to document deployment capability. **Almost none can, because nobody measures it.**

The ceiling moves when the value stops landing on the agency.

A volunteer department has no software budget — so pricing against it caps us at a few thousand dollars a year. The scale unlock is that **staffing reliability is already priced by someone else.**

THE MECHANISM

ISO Public Protection Classification

Every fire district in the country carries an ISO PPC score of 1–10. **Half of that score is department quality, including staffing.** The score sets fire-insurance pricing across the district — directly for commercial property, and through carrier competition for residential.

A department that can *document* and improve staffing reliability is moving a number that prices insurance for every property it covers. We are the only tool that measures the underlying reality.

LINE 1
Agency subscription. ~26,000 volunteer fire + rural EMS agencies. Priced against a grant line, not an operating budget.
LINE 2
District-level contracts. When the deliverable is an ISO-relevant capability record rather than a staffing report, the buyer becomes the district and the county — and the ACV moves from thousands to five figures.
LINE 3
The benchmark dataset. Crew-assembly reliability does not exist as a dataset anywhere today. Aggregated across thousands of departments it is the reference series insurers, state fire marshals and standards bodies have no substitute for.
MOAT
It compounds. Every audit adds to the only benchmark of its kind, which makes the next audit more valuable — a comparison no competitor can reproduce without first collecting the same thousands of exports.

Honest arithmetic: ~26,000 agencies at a \$4-5K subscription is a ~\$120M domestic TAM — a good business, not yet a venture outcome. It becomes one only if line 2 lands and ACV moves toward five figures on the insurance-linked value, or if line 3 becomes a data business in its own right. Both are hypotheses. Neither is provable without the pilot data on the next slide, and we are not going to present them as forecasts.

One agency's spreadsheet changes everything.

We are one dataset away from knowing whether this works. Not a study, not a grant cycle — one CSV from one rural volunteer agency.

01 **Introductions to 3–5 rural EMS agencies or a regional EMS agency.** We run the audit free. Their data proves or kills the thesis inside a month.

02 **The two questions we ask before writing more code:** when a call comes in and no crew forms, what happens and how often — and if you could fund a few staffed hours a week, would you know which hours to pick? If the answers are "often, we don't track it" and "no," we have a company.

03 **A pilot partner willing to co-apply for SIREN funding** using the packet our tool generates — closing the loop from measurement to money.

We would rather find out in March that this doesn't work than in December that we never checked.

— the reason every claim in this deck has a source on it